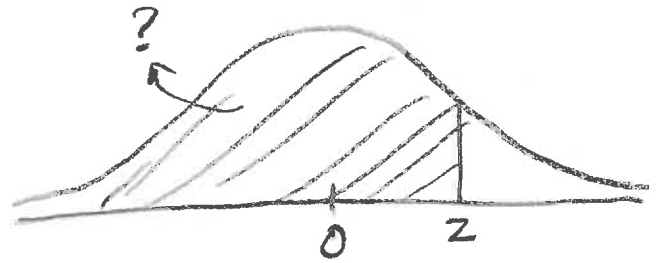
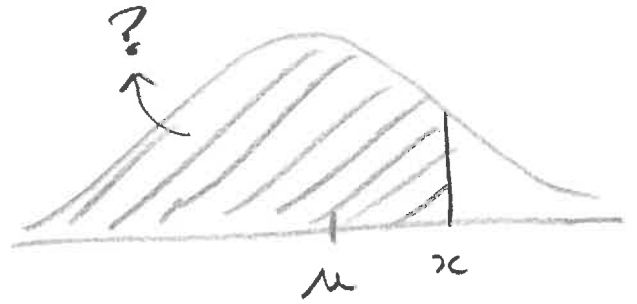


Normal distribution

① Probabilities / Area

$$P(X < x) = ?$$

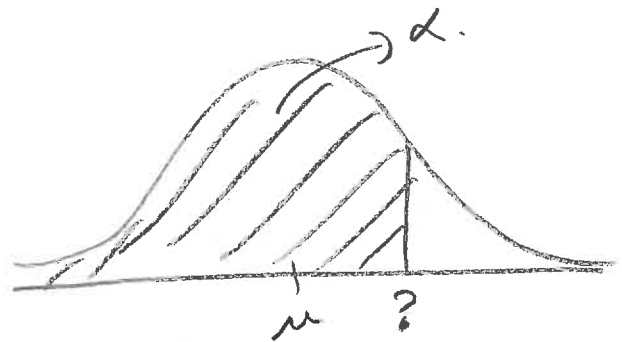
$$\rightarrow P(Z < z) = ?$$



$$\text{Use: } Z = \frac{X - \mu}{\sigma}$$

② Percentiles

$$P(X < ?) = \alpha$$



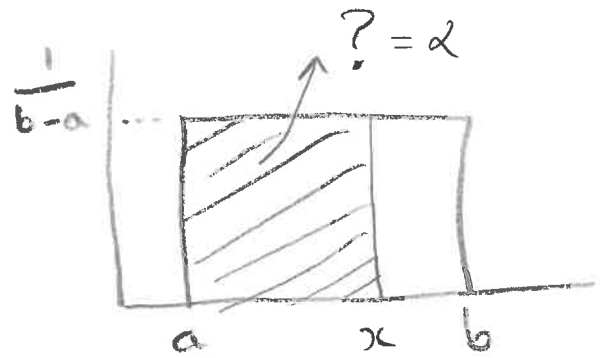
$$\text{Use: } P_{\alpha} = \mu + Z_{\alpha} \sigma$$

$$X = \mu + Z \sigma$$

Uniform distribution

① Probabilities

$$P(X < x) = ?$$

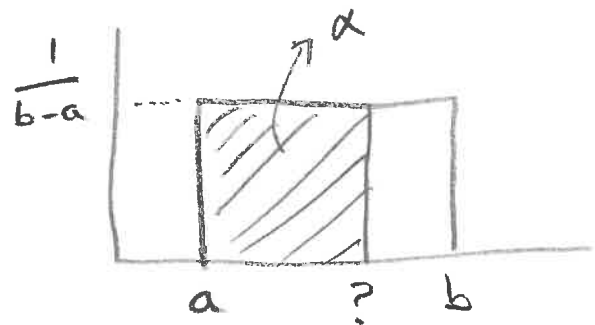


Use: Width $\times$ Height

$$\Rightarrow P(X < x) = \frac{x-a}{b-a} = \alpha$$

② Percentiles

$$P(X < ?) = \alpha$$

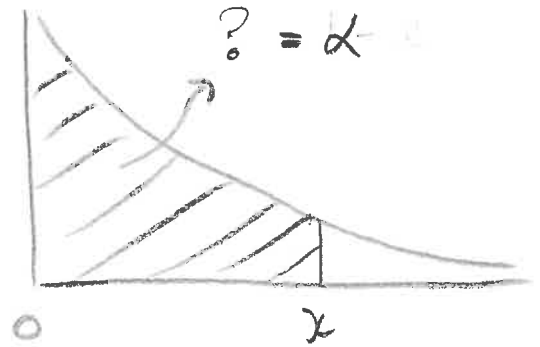


Use: $P_\alpha = a + \alpha(b-a)$

Exponential distribution

① Probabilities

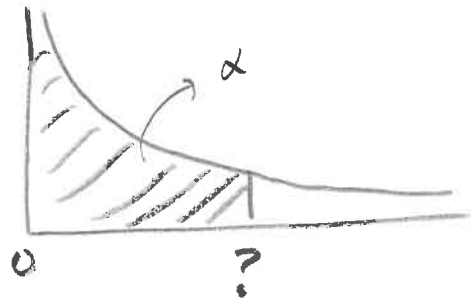
$$P(X < x) = ?$$



Use: $P(X < x) = 1 - e^{-\lambda x}$

② Percentiles

$$P(X < ?) = \alpha$$



Use: $P_\alpha = -\frac{1}{\lambda} \ln(1 - \alpha)$